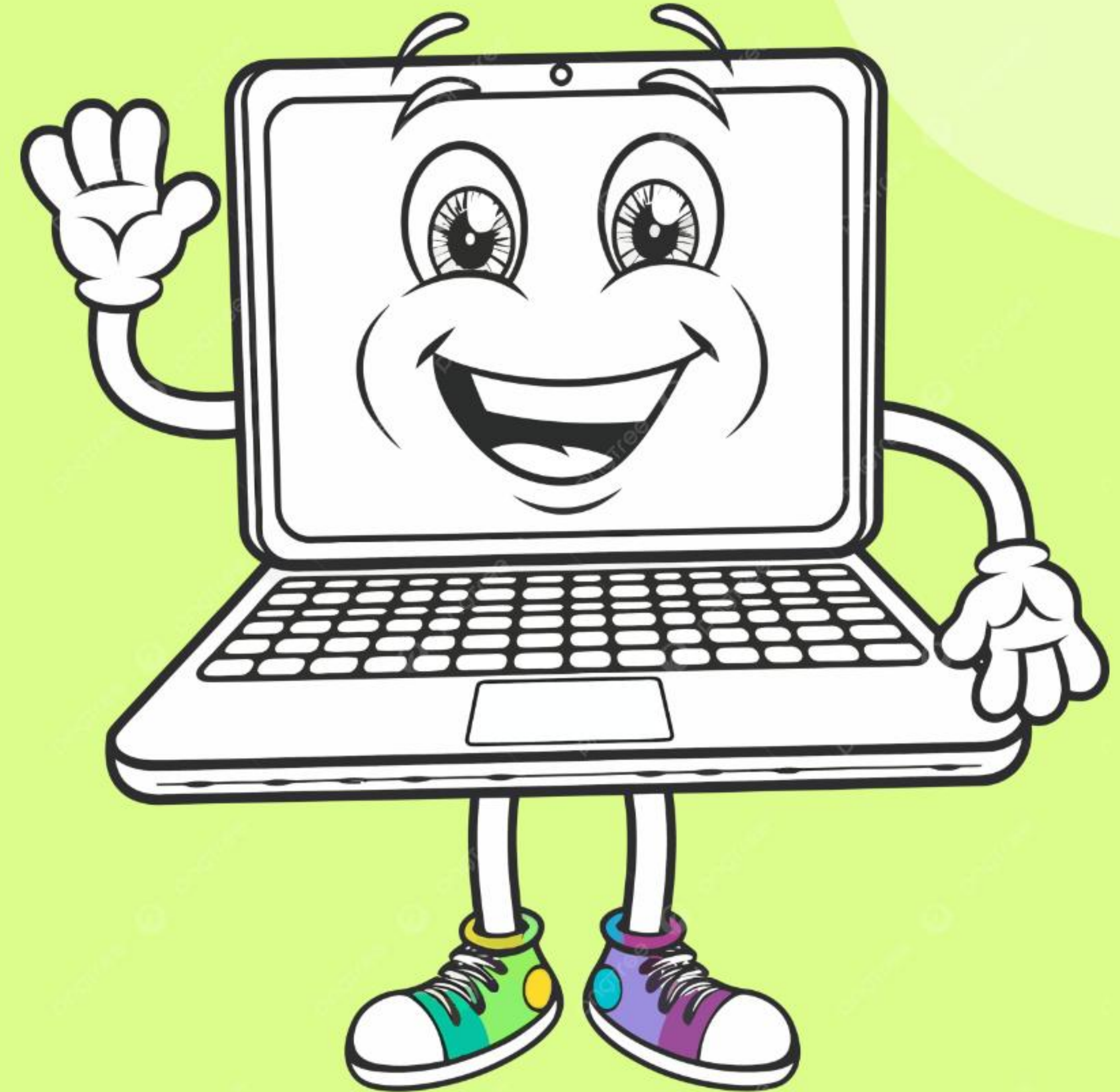


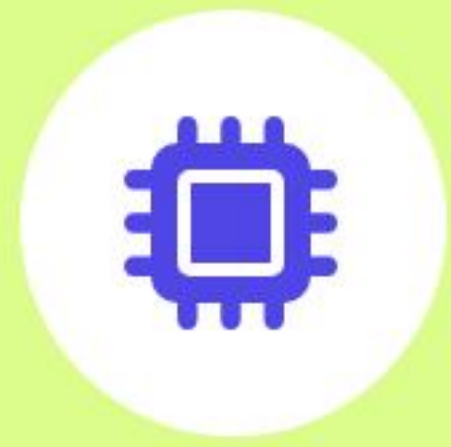
Computer Science

# Hardware Heroes

Grade 4 Discovery Lab



# Hardware vs. Software



## Hardware

The physical parts of the computer that you can **see and touch**.

*Example: Keyboard, Monitor, CPU.*



## Software

Instructions that tell the computer what to do. You **cannot touch it!**

*Example: Games, MS Paint, Windows.*

# Input: Giving Instructions



## Keyboard

Enter data using keys like letters, numbers, and symbols.



## Mouse

A pointing device used to select objects and draw.



## Microphone

Records your voice or sounds into the computer system.

# Output: Seeing Results



**Monitor**

Displays visual information like movies and text.



**Printer**

Gives a "Hard Copy" of your work on paper.



**Speakers**

Lets you hear sounds, music, and voice from your PC.

# Workout: Which device should I use?

1. Movie on a large screen?

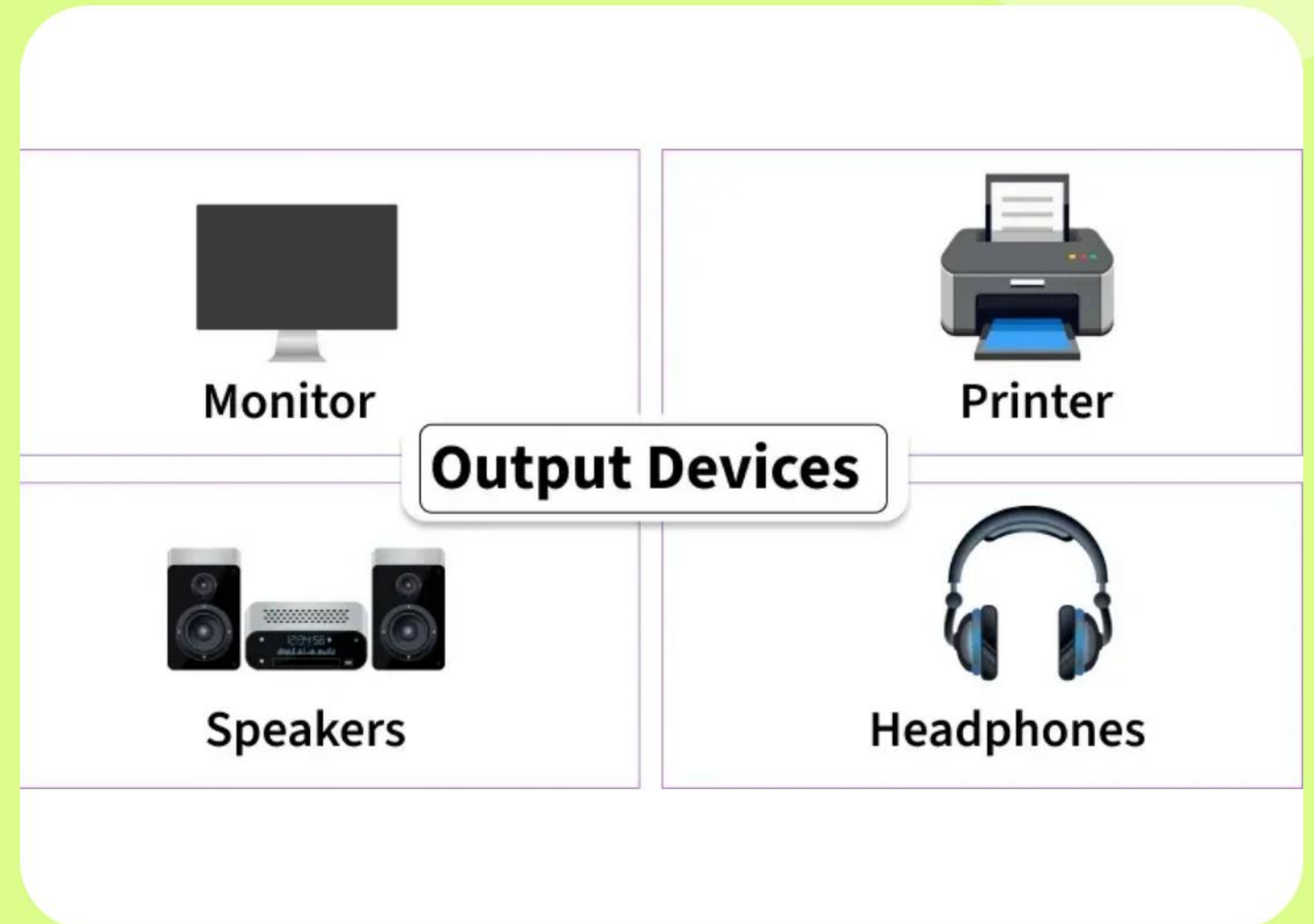
→ Use a **PROJECTOR**

2. Print lots of text in high quality?

→ Use a **LASER PRINTER**

3. Listen to music without disturbing anyone?

→ Use **HEADPHONES**



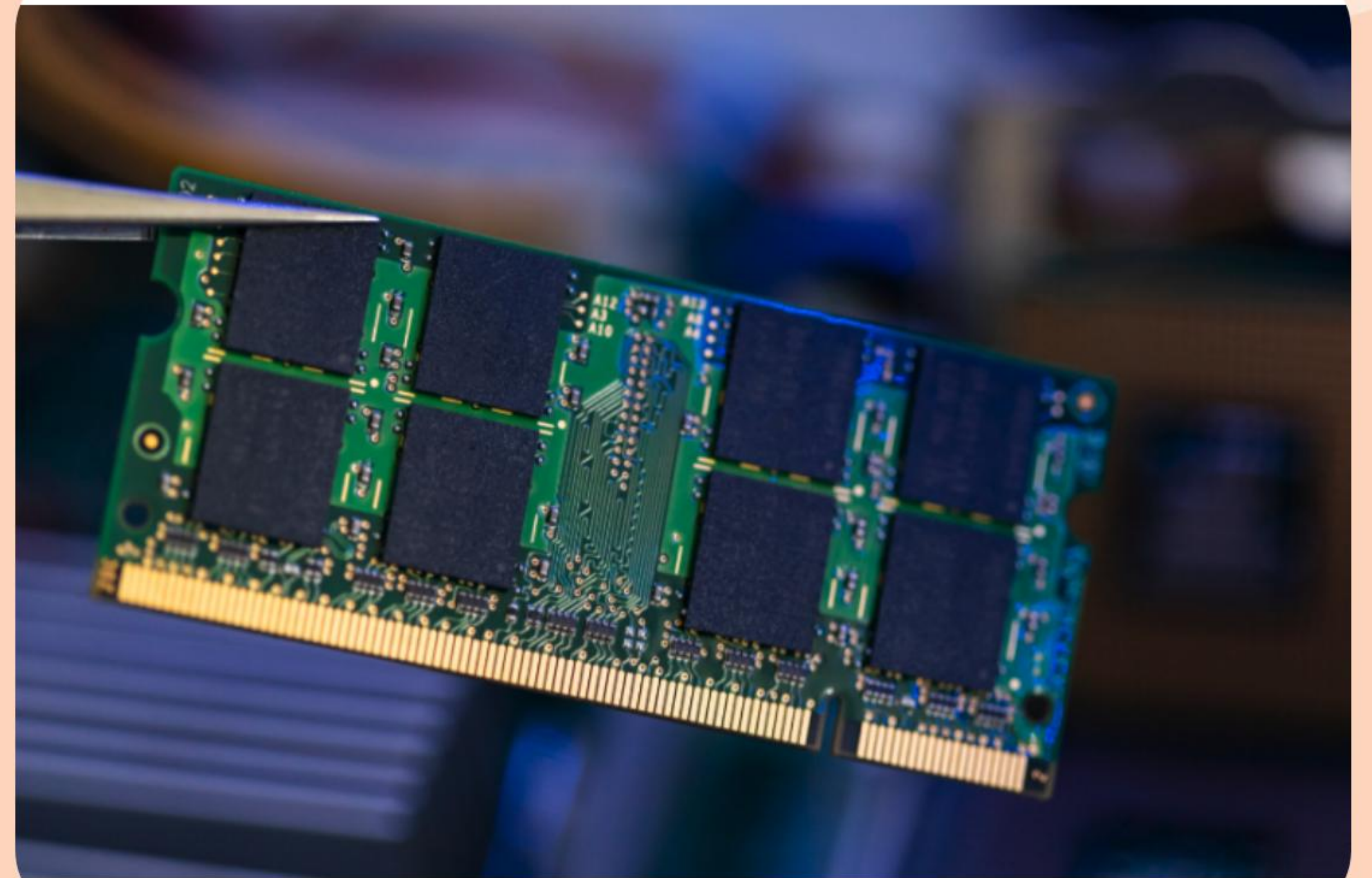
# What is Computer Memory?

Just like you remember your birthday, a computer has **Memory** to store information.

It can store things **Temporarily** (short time) or **Permanently** (long time).

## Two Types:

- Internal Memory (Inside)
- External Memory (Outside)



# Internal Memory: RAM and ROM



**RAM**

**Random Access Memory**

Stores data temporarily while you work. It is erased when the computer is turned off!



**ROM**

**Read Only Memory**

Stores data permanently. It holds start-up instructions and is NOT erased when you turn it off.

# External Memory

This is also called **Secondary Memory**.

We use it to store information for a long time. It is not erased even when the power goes off!

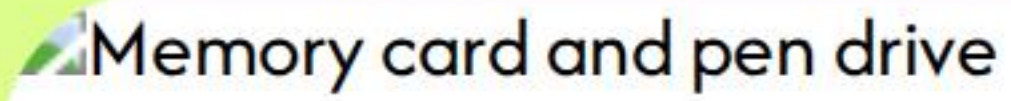
These are called **Storage Devices**.



## Examples:

- Hard Disk (Inside CPU box)
- Pen Drive
- CD / DVD
- Memory Card

# Super Storage: Memory Card

Memory card and pen drive

## The Tiny Hero

A **Memory Card** is very small but can store many photos and videos.

### Commonly used in:

- Digital Cameras
- Smartphones
- Tablets

It is portable, so you can take your data anywhere!

# Quick Look: Storage Devices

| Device    | Type               | Where is it?           |
|-----------|--------------------|------------------------|
| Hard Disk | External/Secondary | Inside the CPU box     |
| Pen Drive | External/Secondary | Plugged into USB port  |
| CD/DVD    | External/Secondary | Inserted in Disk Drive |
| RAM       | Internal/Primary   | On the Motherboard     |

# Quiz Time! (1-3)

**Q1**

Which device is used to watch movies on a big screen?

Check Answer

**Q2**

Is a Mouse Hardware or Software?

Check Answer

**Q3**

Which internal memory gets erased when power goes off?

Check Answer

# Quiz Time! (4-6)

**Q4**

What is also known as  
"Secondary Memory"?

Check Answer

**Q5**

What device gives you a "Hard  
Copy"?

Check Answer

**Q6**

Is ROM erased when you turn off  
the computer?

Check Answer

# Quiz Time! (7-9)

**Q7**

What tiny storage is used in digital cameras?

Check Answer

**Q8**

Which printer gives high-quality text prints?

Check Answer

**Q9**

Which hardware lets you listen to music alone?

Check Answer

# Great Job! Keep Learning!

Explore the technology around you every day! 🚀



# Image Sources



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